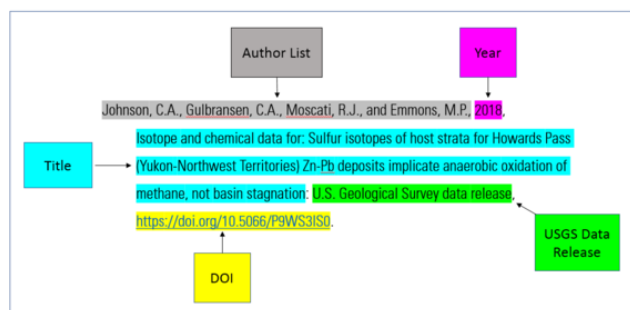


Data Citation

By [Data Management](#)

Data citation is the practice of referencing data products used in research. A data citation includes key descriptive information about the data, such as the title, source, and responsible parties.

Cite Your Own Data, Too!



It is important to cite other people's data when you use them, but it is also important to cite your own data to enable readers to locate and potentially reuse your data.

Expand Image

Digital Object Identifiers in Data Citations



Assigning a digital object identifier to a dataset is important for ensuring that a data citation remains persistent.

Learn about DOIs

Was this page helpful?

Table of Contents

[Why Cite Data?](#)

[Data Citation Elements](#)

[Examples](#)

[Data Citation FAQs](#)

[What the U.S. Geological Survey Manual
Requires](#)

[Recommended Viewing](#)

[References](#)

Why Cite Data?

The goal of data citation is to provide scientific transparency and attribution. Data citations benefit the researcher, funding organization, data repositories, scientific community, and general public. Data citations serve many purposes:

- To aid scientific trustworthiness and reproducibility
- To provide fair credit for data creators or authors, data stewards
- To ensure scientific transparency and reasonable accountability for data authors and stewards;
- To aid in tracking the impact of a dataset and the associated repository
- To help data authors verify how their data are being used
- To help future users identify how others have used the data. (ESIP Data Preservation and Stewardship Committee, 2019)

For more information see [Data Citation FAQ 1.2](#).

Data Citation Elements

A typical data citation generally consists of seven elements. **Was this page helpful?**

- Author
- Publication year
- Title of the data [see [Metadata Creation: Best Practices for Metadata Creation](#) for guidelines]
- Version ID [see [FSP Versioning Guidance](#) for more information]
- Publisher
- Digital Object Identifier [see [Publish/Share > Digital Object Identifiers](#) for more information]
- Access date and time

If relevant, other elements can be included such as query parameters, direct access link, data format, 3rd party producer, name of editor or contributor, publication place, data within a larger work.

For more details on the elements that should be included in a data citation see [Data Citation FAQ 1.3](#).

Examples

Example Data Citations in a Publication's Text and References

- Reference to data release in publication: <https://doi.org/10.3133/sir20225004>
 - *In text*: “Ground surface elevation (GSE) data were collected at the downstream section of the Middle Rice Creek Restoration (fig. 1; Groten and others, 2022)”.
 - *In references*: Groten, J.T., Livdahl, C.T., and DeLong, S.B., 2022, Suspended sediment and bedload data, simple linear regression models, loads, elevation data, and FaSTMECH models for Rice Creek, Minnesota, 2010–2019: U.S. Geological Survey data release, <https://doi.org/10.5066/P9SJIY32>.
- Reference to data release in publication: <https://doi.org/10.3133/sir20225013>.
 - *In text*: “Scripts and data used to perform analyses for this study are available in a USGS data release (Tatge and others, 2022)”.
 - *In references*: Tatge, W.S., Nustad, R.A., and Galloway, J.M., 2022, Data and scripts used in water–quality trend and load analysis in the Heart River Basin, North Dakota, 1970–2020: U.S. Geological Survey data release, access February 2022 at <https://doi.org/10.5066/P987APZ8>.
- Reference to data release in publication: <https://doi.org/10.3133/sir20225004> **Was this page helpful?**

[j.isprsjprs.2021.08.014.](#)

- *In text*: “Model predictor variables and outputs developed for this study are included in Enwright et al. (2021)”.
- *In references*: Enwright, N.M., Kranenburg, C.J., Patton, B.A., Dalyander, P.S., Brown, J.A., Piazza, S.C., Cheney, W.C., 2021, Developing Bare-Earth Digital Elevation Models from Structure-from-Motion Data on Barrier Islands: U.S. Geological Survey data release, <https://doi.org/10.5066/P99PX003>.

For examples of the types of data you should cite in reports and publications, see [Data Citation FAQ 2.1](#).

Example Data Citations for USGS Released Data

- Groten, J.T., Livdahl, C.T., and DeLong, S.B., 2022, Suspended sediment and bedload data, simple linear regression models, loads, elevation data, and FaSTMECH models for Rice Creek, Minnesota, 2010–2019: U.S. Geological Survey data release, <https://doi.org/10.5066/P9SJIY32>.
- Abdollahian, N., Jones, J.L., Ball, J.L., Wood, N.J., and Mangan, M.T., 2018, Data release for results of societal exposure to California's volcanic hazards (ver. 3.0, November 2019): U.S. Geological Survey data release, accessed February 10, 2020, at <https://doi.org/10.5066/F7W66JRH>.
- U.S. Geological Survey, 2020, BioData—Aquatic bioassessment data for the Nation: U.S. Geological Survey database, accessed February 20, 2020, at <https://doi.org/10.5066/F77W698B>.

Example Data Citations for Non-USGS Data

- Hall, D.K., and G. A. Riggs, 2016, MODIS/Terra snow cover daily L3 global 500m grid (ver. 6.0): National Snow and Ice Data Center Data Set MOD10A1, accessed February 2, 2019, at <https://doi.org/10.5067/MODIS/MOD10A1.006>. [Query parameters: Oct. 2007– Sep. 2008, 84°N, 75°W; 44°N, 10°W].
- Ocean Networks Canada Society, 2019, Barkley Canyon upper slope fluorometer turbidity deployed 2019-05-16: Ocean Networks Canada Society dataset, accessed April 13, 2020, at <https://doi.org/10.34943/fa04d675-3df2-4dc3-810b-cb365f7ec492>. [Subset Query: 8297991].

For more examples of data citations see [Data Citation FAQ 1.4](#).

Still have questions?

Was this page helpful?

USGS has documented [frequently asked questions about data citation](#), including:

- Differences in citing static, versioned, and dynamic data
- Who to include in the author list and when it is appropriate for the author list to change over time
- How to cite a data product without a recommended data citation

What the *U.S. Geological Survey Manual* Requires:

The USGS Survey Manual chapter [SM 502.8 Fundamental Science Practices: Review and Approval of Scientific Data for Release](#) requires that data approved for release must be assigned a persistent identifier, specifically a Digital Object Identifier (DOI) for scientific data obtained from the USGS registration agent, and be accompanied by a recommended citation.

[SM 1100.5 - Authorship, Acknowledgments, and Credits in USGS Information Products](#) states that authorship of USGS information products provides credit and assigns responsibility for information contained in the product. The senior author is responsible for acknowledging contributions and crediting cooperators from other agencies.

Recommended Viewing

- [Data Citation Tips from NASA's Land Processes DAAC](#)

References

- ESIP Data Preservation and Stewardship Committee, 2019, Data citation guidelines for earth science data, version 2: Earth Science Information Partners web page, accessed April 22, 2020, at <https://doi.org/10.6084/m9.figshare.8441816>.

[Privacy Policy](#)

[Site Map](#)

[DOI and USGS link policies apply](#)

[No FEAR Act](#)

[USA.gov](#)

[Vulnerability Disclosure](#)

[Careers and Employment](#)

[Doing Business](#)

[Emergency Management](#)



U.S. Geological Survey
U.S. Department of the Interior

Contact USGS

answers.usgs.gov

Was this page helpful?